Aknur Sherkhan

San Francisco, CA · aknur@uni.minerva.edu · Linkedin · GitHub · Portfolio

Education

Minerva University — **San Francisco, CA**

Candidate for Bachelor's in Computational and Social Sciences (Data Science and Economics) — *Expected Graduation May 2027*

- Relevant coursework: Machine Learning, Simulation & Modeling. Data Structures & Algorithms, Bayesian Statistics, Probability & Statistics, Econometrics, Linear Algebra, Multivariable Calculus.

Experience

Minerva University — **San Francisco, CA**

Data Analysis Intern - Admissions Team — *Sep 2024 – Present*

- Analyzed 20,000+ applicant records using Python (Pandas, NumPy) to identify trends in applicant quality and volume; built segmentation logic that directly guided team prioritization during peak review cycles.
- Developed Python-based QA scripts to resolve data inconsistencies across 3000+ applications, maintaining high data integrity.

Puerta 18 — **Buenos Aires, Argentina**

CRM and Data Systems Developer — *Sep 2025 - Apr 2026*

- Built the organization's first data infrastructure from scratch: digitized 18 years of unstructured records into a centralized PostgreSQL system; defined data standards and documentation for long-term maintainability.
- Engineered automated reporting pipelines (Python, SQL) and live KPI dashboards used daily by staff; eliminated 15+ hrs/week of manual reporting and increased admin capacity by 0.4 FTE.

Terra Education at UC Berkeley — **Berkeley, CA**

Data Science and ML Teaching Assistant — *Jun 2025 - Aug 2025*

- Guided 150+ high school students through end-to-end DS/ML projects: data cleaning, feature engineering, model training and evaluation; coached teams on scoping ambiguous problems and translating model outputs into defensible decisions.

GSP Cloud — **Seoul, South Korea**

Product Development Intern — *Jan 2025 - May 2025*

- Designed and executed A/B tests across ~400 users; built behavioral log analysis pipeline to identify engagement drop-off patterns, driving a 12% increase in user retention.
- Evaluated GPT-4o vs. GPT-4 on 50+ test prompts; documented regression-test protocol for ongoing model monitoring.

Minerva University — **San Francisco, CA**

Student Research Intern at Psychology Lab — *Sep 2023 – May 2024*

- Processed and visualized multi-site experimental datasets (Pandas, Matplotlib) for studies with Ivy League partner institutions.
- Improved fraud-prevention workflows by identifying anomalous participant patterns on Lookit (MIT's behavioral research platform), increasing data integrity across 450+ studies.

Projects

Argentine Football Attendance Analysis | *Python, PyMC, ArviZ, pandas*

- Built two competing Bayesian models (complete pooling vs. hierarchical partial pooling) on 240 Argentine Primera División games; justified Negative Binomial likelihood over Poisson via overdispersion analysis.
- Resolved MCMC divergences via non-centered reparametrization; PSIS-LOO gave 100% stacking weight to the hierarchical model; imputed 22 missing games with 89% HDI uncertainty intervals.

Urban Traffic Congestion Simulation | *Python, NumPy, OpenStreetMap, Monte Carlo, Network Analysis*

- Built an OOP traffic simulator using real OpenStreetMap road networks; modeled congestion via the Greenshields speed-density model with dynamic rerouting across 50-run Monte Carlo evaluations.
- Identified edge betweenness centrality as a dominant structural predictor of congestion (Spearman $\rho = 0.93$), validated across Berlin and Buenos Aires road networks.

Imperium Volunteers | *Statistical Analyst*

- Modeled volunteer engagement and dropout risk using Python (pandas, scikit-learn) on 500+ records; improved early identification of low-engagement volunteers by 30%, contributing to a 15% increase in participation.

Personalized Photo Recommendation Model | *CLIP, KMeans, Cosine similarity*

- Framed photo selection as a revealed preference problem; used CLIP embeddings and cosine similarity to rank camera-roll photos by predicted posting likelihood, achieving 3x Precision@10 improvement over baseline.

Skills

- **Languages:** Python (advanced), SQL (advanced), R (intermediate), JavaScript/React (intermediate)
- **Data & Analysis:** Pandas, NumPy, PyMC, ArviZ, Matplotlib, Tableau, scikit-learn, Excel/Sheets (pivot tables, index/match)
- **ML & Modeling:** Bayesian modeling, experiment design, causal inference, PyTorch, TensorFlow, LLM integration